

Vectors: Components and Operations

ANSWER KEY



Vector arithmetic

- 1 Let $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 4 \rangle$. Find:

a $\mathbf{u} + \mathbf{v} = \langle 2, 2 \rangle$

b $3\mathbf{u} - 2\mathbf{v} = \langle 11, -14 \rangle$

c $|\mathbf{u}| = \sqrt{13}$

- 2 Find the component form and magnitude of the vector from $P(1, 3)$ to $Q(5, -2)$.

$\vec{PQ} = \langle 4, -5 \rangle$, $|\vec{PQ}| = \sqrt{16 + 25} = \sqrt{41}$.

- 3 Find the unit vector in the direction of $\mathbf{v} = \langle 6, -8 \rangle$.

$|\mathbf{v}| = 10$, so $\hat{\mathbf{v}} = \langle \frac{3}{5}, -\frac{4}{5} \rangle$.

- 4 Write the vector of magnitude 12 making an angle of 150° with the positive x-axis in component form (exact values).

$\langle 12\cos 150^\circ, 12\sin 150^\circ \rangle = \langle -6\sqrt{3}, 6 \rangle$.

Applications

- 5 A plane flies due east at 200 mph while the wind blows due north at 50 mph.

a Find the magnitude of the plane's resultant velocity, to one decimal place.

$\sqrt{200^2 + 50^2} = \sqrt{42500} \approx 206.2$ mph.

b Find the direction of the resultant, as an angle north of east to two decimal places.

$\theta = \tan^{-1}\left(\frac{50}{200}\right) \approx 14.04^\circ$ north of east.

- 6 Two forces $\mathbf{F}_1 = \langle 30, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 40 \rangle$ N act on an object. Find the magnitude of the resultant force.

$|\mathbf{F}_1 + \mathbf{F}_2| = |\langle 30, 40 \rangle| = 50$ N.

- 7 A boat aims to cross a river by heading due north at 4 m/s, but the current pushes it east at 1.5 m/s.

a Find the boat's resultant speed, to two decimal places. $\sqrt{4^2 + 1.5^2} = \sqrt{18.25} \approx 4.27$ m/s.

b If the river is 200 m wide, find how long the crossing takes. Only the northward component moves it across: $t = \frac{200}{4} = 50$ s.

c Find how far downstream (east) the boat lands. Drift = $1.5 \times 50 = 75$ m.

Position vectors and geometry

- 8 Given $A(-2, 5)$ and $B(4, -3)$, find the midpoint of $\overline{AB}$ and the vector $\vec{AB}$.

Midpoint: $\left(\frac{-2+4}{2}, \frac{5-3}{2}\right) = (1, 1)$. $\vec{AB} = \langle 6, -8 \rangle$.

- 9 Let $\mathbf{a} = \langle 2, 3 \rangle$ and $\mathbf{b} = \langle -4, 1 \rangle$. Find a vector $\mathbf{c}$ such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \langle 0, 0 \rangle$, and verify your answer.
 $\mathbf{c} = -(\mathbf{a} + \mathbf{b}) = -\langle -2, 4 \rangle = \langle 2, -4 \rangle$. Check: $\langle 2, 3 \rangle + \langle -4, 1 \rangle + \langle 2, -4 \rangle = \langle 0, 0 \rangle$. ✓
- 10 Express $\mathbf{v} = \langle 5, -12 \rangle$ in terms of the standard unit vectors $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{v}|$.
 $\mathbf{v} = 5\mathbf{i} - 12\mathbf{j}$. $|\mathbf{v}| = \sqrt{25 + 144} = 13$.
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More applications

- 11 An airplane has an airspeed of 305 mph and flies on a heading 7.5° west of due north, while a 40 mph wind blows due east. Find the plane's resultant ground velocity vector, and its ground speed, to one decimal place (take east as the positive x -direction and north as the positive y -direction).
Plane's own velocity: $\langle -305\sin 7.5^\circ, 305\cos 7.5^\circ \rangle \approx \langle -39.8, 302.4 \rangle$. Adding the wind $\langle 40, 0 \rangle$:
resultant $\approx \langle 0.2, 302.4 \rangle$. Ground speed $\approx \sqrt{0.2^2 + 302.4^2} \approx 302.4$ mph — very nearly due north, as intended.
- 12 Given $\mathbf{u} = \langle 4, -1 \rangle$, find a vector orthogonal to $\mathbf{u}$ with the same magnitude as $\mathbf{u}$.
Swap components and negate one: $\langle 1, 4 \rangle$. Check: $\mathbf{u} \cdot \langle 1, 4 \rangle = 4 - 4 = 0$ (orthogonal), and $|\langle 1, 4 \rangle| = \sqrt{17} = |\mathbf{u}|$.